



线性代数第一次作业

$$3(1) \begin{vmatrix} 10 & 2 & 8 \\ 15 & 3 & 12 \\ 20 & 12 & 32 \end{vmatrix} = 2 \times 3 \times 4 \begin{vmatrix} 5 & 1 & 4 \\ 5 & 1 & 4 \\ 5 & 3 & 8 \end{vmatrix} = 0$$

$$12) \begin{vmatrix} 0 & 0 & 0 & 4 \\ 0 & 0 & 4 & 3 \\ 0 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{vmatrix} = -4 \cdot \begin{vmatrix} 0 & 0 & 4 \\ 0 & 4 & 3 \\ 4 & 3 & 2 \end{vmatrix} = \dots = (-4) \times 4 \times (-4) \times 4 = 256$$

$$13) \begin{vmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ 0 & b & a & 0 \\ b & 0 & 0 & a \end{vmatrix} = \begin{vmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ 0 & b & a & 0 \\ b & 0 & 0 & a \end{vmatrix} = \begin{vmatrix} a^2 & b^2 \\ b^2 & a^2 \end{vmatrix} = a^4 - b^4$$

$$14) \begin{vmatrix} 5 & 0 & 4 & 2 \\ 1 & -1 & 2 & 1 \\ 4 & 1 & 2 & 0 \\ 1 & 1 & 1 & 1 \end{vmatrix} \xrightarrow{r_1 - r_3 - r_4} \begin{vmatrix} 0 & 0 & 0 & 1 \\ 1 & -1 & 2 & 1 \\ 4 & 1 & 2 & 0 \\ 1 & 1 & 1 & 1 \end{vmatrix} = - \begin{vmatrix} 1 & -1 & 2 \\ 4 & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix} \xrightarrow{\substack{r_1 - r_2 \\ r_2 - r_3}} \begin{vmatrix} -3 & -2 & 0 \\ 3 & 0 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= 3 \times \begin{vmatrix} -2 & 0 \\ 1 & 1 \end{vmatrix} + 1 \times \begin{vmatrix} -3 & -2 \\ 1 & 1 \end{vmatrix} = -6 + (-1) = -7$$

$$15) \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 2 & 3 \end{vmatrix} \xrightarrow{r_3 - r_1 - r_2} \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 0 & -1 & -5 & -4 \\ 4 & 1 & 2 & 3 \end{vmatrix} \xrightarrow{r_2 - 2r_1} \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -7 \\ 0 & -1 & -5 & -4 \\ 4 & 1 & 2 & 3 \end{vmatrix}$$

$$\xrightarrow{r_3 - r_2} \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -7 \\ 0 & 0 & -3 & 3 \\ 4 & 1 & 2 & 3 \end{vmatrix} \xrightarrow{c_3 + c_4} \begin{vmatrix} 1 & 2 & 7 & 4 \\ 0 & -1 & -9 & -7 \\ 0 & 0 & 0 & 3 \\ 4 & 1 & 5 & 3 \end{vmatrix} = -3 \times \begin{vmatrix} 1 & 2 & 7 \\ 0 & -1 & -9 \\ 4 & 1 & 5 \end{vmatrix}$$

$$= -3 \times (-5 + (-22) - (-28) - (-9)) = -3 \times (-40) = 120$$

$$(b) \begin{vmatrix} a & 0 & b & 0 \\ 0 & c & 0 & d \\ x & 0 & y & 0 \\ 0 & u & 0 & v \end{vmatrix} \xrightarrow{r_3 - \frac{x}{a}r_1} \begin{vmatrix} a & 0 & b & 0 \\ 0 & c & 0 & d \\ 0 & 0 & y - \frac{bx}{a} & 0 \\ 0 & u & 0 & v \end{vmatrix} \xrightarrow{r_4 - \frac{u}{c}r_2} \begin{vmatrix} a & 0 & b & 0 \\ 0 & c & 0 & d \\ 0 & 0 & y - \frac{bx}{a} & 0 \\ 0 & 0 & 0 & v - \frac{ud}{c} \end{vmatrix}$$

$$= ac(y - \frac{bx}{a})(v - \frac{ud}{c}) = (ay - bx)(cv - ud) \neq 0$$

(第(3)问做错)。不能直接切块。下面重做:

$$(3) \begin{vmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ 0 & b & a & 0 \\ b & 0 & 0 & a \end{vmatrix} \xrightarrow{r_2 - \frac{b}{a}r_1} \begin{vmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ 0 & 0 & a - \frac{b^2}{a} & b \\ b & 0 & 0 & a \end{vmatrix} \xrightarrow{r_4 - \frac{b}{a}r_1} \begin{vmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ 0 & 0 & a - \frac{b^2}{a} & b \\ 0 & 0 & -\frac{b^2}{a} & a \end{vmatrix}$$

$$\xrightarrow{r_4 + \frac{b^2}{a^2}r_3} \begin{vmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ 0 & 0 & a - \frac{b^2}{a} & b \\ 0 & 0 & 0 & a - \frac{b^4}{a^3} \end{vmatrix} = a \cdot a \cdot a \cdot \left(a - \frac{b^4}{a^3}\right) = a^4 - b^4$$

$$4(1) \text{ 解: } \begin{vmatrix} a^2 & ab & b^2 \\ 2a & a+b & 2b \\ 1 & 1 & 1 \end{vmatrix} = a^2(ab) + 2ab^2 + 2ab^2 - b^2(ab) - 2a^2b - 2a^2b$$

$$= a^3 + 3ab^2 - 3a^2b - b^3 = (a-b)^3$$

$$(2) \text{ 解: } \begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix} \xrightarrow{r_3 - r_2} \begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ 0 & -1 & 1 \end{vmatrix} \xrightarrow{r_2 - r_1} \begin{vmatrix} 1+a & b & c \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix}$$

$$= (-1) \times (-1)^5 \begin{vmatrix} 1+a & c \\ -1 & 0 \end{vmatrix} + 1 \times (-1)^6 \begin{vmatrix} 1+a & b \\ -1 & 1 \end{vmatrix} = c + 1 + a + b = 1 + a + b + c$$

$$(3) \text{ 解: } \begin{vmatrix} 1 & ax & a^2+x^2 \\ 1 & ay & a^2+y^2 \\ 1 & az & a^2+z^2 \end{vmatrix} \xrightarrow{r_3 - r_2} \begin{vmatrix} 1 & ax & a^2+x^2 \\ 1 & ay & a^2+y^2 \\ 0 & a(z-y) & z^2-y^2 \end{vmatrix} \xrightarrow{r_2 - r_1} \begin{vmatrix} 1 & ax & a^2+x^2 \\ 0 & a(y-x) & y^2-x^2 \\ 0 & a(y-z) & y^2-z^2 \end{vmatrix}$$

$$= \begin{vmatrix} a(y-x) & y^2-x^2 \\ a(y-z) & z^2-y^2 \end{vmatrix} = (y-x)(z-y) \begin{vmatrix} a & y+x \\ a & y+z \end{vmatrix} = a(x-y)(y-z)(z-x)$$

$$(4) \text{ 解: } D_n = \begin{vmatrix} \lambda & -1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & -1 & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & & & & \lambda & -1 \\ a_n & a_{n-1} & \cdots & \cdots & \lambda + a_1 \end{vmatrix} = \lambda \begin{vmatrix} \lambda & -1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & -1 & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & & & & \lambda & -1 \\ a_{n-1} & \cdots & \cdots & \lambda + a_1 \end{vmatrix} + (-1)^{n+1} \begin{vmatrix} \lambda & -1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & -1 & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & & & & \lambda & -1 \\ 0 & \cdots & \cdots & \lambda & -1 \end{vmatrix} \cdot a_n$$

$$= \lambda D_{n-1} + (-1)^{n+1} \times (-1)^{n-2} a_n = \lambda D_{n-1} + a_n$$

$$= \lambda (\lambda D_{n-2} + a_{n-1}) + a_n = \lambda^2 D_{n-2} + \lambda a_{n-1} + a_n = \cdots$$

$$= \lambda^n + a_1 \lambda^{n-1} + \cdots + a_{n-1} \lambda + a_n$$

$$\text{51 解: } D_n = (a+b) D_{n-1} + (-1)^3 \cdot ab \cdot (a+b)^{n-2}$$

$$= (a+b) D_{n-1} - ab \cdot x/x \cdot D_{n-2}$$

$$= (a+b) D_{n-1} - ab D_{n-2}$$

$$\text{特征方程为 } x^2 - (a+b)x + ab = 0 \Rightarrow x_1 = a, x_2 = b$$

$$\therefore D_n = A a^n + B b^n, D_1 = a+b, D_2 = (a+b)^2 - ab = \frac{a^3 - b^3}{a-b}$$

$$\text{解得 } A = \frac{a}{a-b}, B = \frac{-b}{a-b} \therefore D_n = \frac{a^{n+1} - b^{n+1}}{a-b}$$

$$\text{51 解: } \begin{vmatrix} 1+x & 2 & 3 \\ 7 & 2+x & 3 \\ 1 & 2 & 3+x \end{vmatrix} \xrightarrow{r_1-r_2} \begin{vmatrix} x-x & 0 & 0 \\ 1 & 2+x & 3 \\ 1 & 2 & 3+x \end{vmatrix} \xrightarrow{c_2+c_1} \begin{vmatrix} x & 0 & 0 \\ 1 & 3+x & 3 \\ 1 & 3 & 3+x \end{vmatrix}$$

$$= x \cdot \begin{vmatrix} 3+x & 3 \\ 3 & 3+x \end{vmatrix} = ((x+3)^2 - 9) \cdot x = 0 \Rightarrow \boxed{x_1 = 0, x_2 = -6}$$

$$\text{11 解: } \begin{vmatrix} x & a & b & c \\ a & x & c & b \\ b & c & x & a \\ c & b & a & x \end{vmatrix} = \begin{vmatrix} x+a+b+c & x+atb+c & x+atb+c & x+atb+c \\ a & x & c & b \\ b & c & x & a \\ c & b & a & x \end{vmatrix}$$

$$= (x+a+b+c) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & x & c & b \\ b & c & x & a \\ c & b & a & x \end{vmatrix} = \cancel{(x+a+b+c)} \cdot \cancel{\begin{vmatrix} x & c & b \\ c & x & a \\ b & a & x \end{vmatrix}} + \cancel{\begin{vmatrix} a & c & b \\ b & x & a \\ c & a & x \end{vmatrix}}$$

$$\cancel{\begin{vmatrix} a & x & b \\ b & c & a \\ c & b & x \end{vmatrix}} + \cancel{\begin{vmatrix} a & x & c \\ b & c & x \\ c & b & a \end{vmatrix}}$$

$$\therefore x_1 = -(a+b+c)$$

$$\text{同理, } r_1+r_2 - (c_3+c_4) \text{ 得一行均为 } x+a-b-c, \text{ 故 } x_2 = -a+b+c$$

$$r_1 - r_2 + r_3 - r_4 \text{ 得一行均为 } x-a+b-c, \text{ 故 } x_3 = a-b+c$$

$$r_1 - r_2 - r_3 + r_4 \text{ 得一行均为 } x-a-b+c, \text{ 故 } x_4 = a+b-c$$

观察行列式, 知展开后为关于 x 的四次方程 (x^4 前系数不为 0)

故有不超过 4 个实根。即我们找到的 4 个。

$$\therefore \boxed{x_1 = -a-b-c, x_2 = -a+b+c, x_3 = a-b+c, x_4 = a+b-c}$$

5(3) 由范德蒙德行列式: (记 $x_1=1, x_2=-1, x_3=2, x_4=x$)

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & (-1) & 2 & x \\ 1^2 & (-1)^2 & 2^2 & x^2 \\ 1^3 & (-1)^3 & 2^3 & x^3 \end{vmatrix} = \prod_{1 \leq i < j \leq 4} (x_j - x_i) = 0$$

$\Rightarrow$ (去掉常数项, 只考虑含 x 的项), $(x-1)(x+1)(x-2) = 0$

$$\therefore x_1 = 1, x_2 = -1, x_3 = 2$$

6(III) 解: $A_{32} = (-1)^5 M_{32}$

$$= (-1)^5 \begin{vmatrix} 3 & -1 & 2 \\ -5 & 3 & -4 \\ 1 & 3 & -3 \end{vmatrix} = -(-27 + 4 - 30 - 6 + 36 + 15) = 8$$

或 $A_{31} + 3A_{32} - 2A_{33} + 2A_{34}$ 错用: 把 A_{ij} 当成 M_{ij}

$$= \begin{vmatrix} 3 & 1 & -1 & 2 & c_1 + 3c_2 & 0 & 0 & -1 & 0 \\ -5 & 1 & 3 & 4 & c_2 + c_3 & 4 & 4 & 3 & 10 \\ 1 & 0 & 3 & -2 & c_4 + 2c_3 & -5 & -5 & -2 & -6 \\ 1 & -5 & 3 & -3 & & 10 & -2 & 3 & 3 \end{vmatrix}$$

$$= (-1)^4 \begin{vmatrix} 4 & 4 & 10 \\ -5 & -5 & -6 \\ 10 & -2 & 3 \end{vmatrix} = \begin{vmatrix} 0 & 4 & 10 \\ 0 & 5 & 6 \\ 12 & -2 & 3 \end{vmatrix}$$

$A_{31} + 3A_{32} - 2A_{33} + 2A_{34}$

$$= \begin{vmatrix} 3 & 1 & -1 & 2 \\ -5 & 1 & 3 & 4 \\ 1 & 3 & -2 & 2 \\ 1 & -5 & 3 & -3 \end{vmatrix} \xrightarrow{r_3 + c_4} \begin{vmatrix} 3 & 1 & -1 & 1 \\ -5 & 1 & 3 & 7 \\ 1 & 3 & -2 & 0 \\ 1 & -5 & 3 & 0 \end{vmatrix}$$

$$\xrightarrow{r_2 - 7r_1} \begin{vmatrix} 3 & 1 & -1 & 1 \\ -26 & -6 & 10 & 0 \\ 1 & 3 & -2 & 0 \\ 1 & -5 & 3 & 0 \end{vmatrix} = - \begin{vmatrix} -26 & -6 & 10 \\ 1 & 3 & -2 \\ 1 & -5 & 3 \end{vmatrix} \xrightarrow{r_1 - r_2 + 2r_3} \begin{vmatrix} 0 & -6 & 10 \\ -6 & 3 & -2 \\ 12 & -5 & 3 \end{vmatrix}$$

$$= (-360 - 36 \times 3 + 144 + 300) \times (-1) = 24$$

$$(1) \text{ 解: } \begin{vmatrix} 1 & x & y & z \\ x & 1 & 0 & 0 \\ y & 0 & 1 & 0 \\ z & 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} - x \begin{vmatrix} x & 0 & 0 \\ y & 1 & 0 \\ z & 0 & 1 \end{vmatrix} + y \begin{vmatrix} x & 1 & 0 \\ y & 0 & 0 \\ z & 0 & 1 \end{vmatrix}$$

$$-z \begin{vmatrix} x & 1 & 0 \\ y & 0 & 1 \\ z & 0 & 0 \end{vmatrix} = 1 - x^2 - y^2 - z^2 = 1 \Rightarrow x^2 + y^2 + z^2 = 0$$

$$\therefore x = y = z = 0$$

$$(2) \text{ 解: } \begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} \xrightarrow{C_1 - C_2} \begin{vmatrix} x-1 & 1 & 1 & 1 \\ x & 1-x & 1 & 1 \\ 0 & 1 & 1+y & 1 \\ 0 & 1 & 1 & 1-y \end{vmatrix}$$

$$\xrightarrow{r_2 - r_1} \begin{vmatrix} x & 1 & 1 & 1 \\ 0 & -x & 0 & 0 \\ 0 & 1 & 1+y & 1 \\ 0 & 1 & 1 & 1-y \end{vmatrix} = x \cdot (-x) \begin{vmatrix} 1+y & 1 \\ 1 & 1-y \end{vmatrix}$$

$$= -x^2 \cdot (1-y^2-1) = x^2 y^2$$

Ch1-1 作业分析

作业：习题一：3,4,5,6

*少数同学行列式直接计算没有简化，另外也有一些同学计算错误，还有极少数同学解题缺少过程。

**少数同学简化行列式时用到参数做分母，如 3(3)用 $r_3-(b/a)r_1$ 、3(6)用 $r_3-(x/a)r_1$ 、4(4)用 $c_2+(1/\lambda)c_1$ 、6(3)用 $c_1-c_2, c_2-(1/x)c_1, c_3-(1/x)c_1, c_4-(1/x)c_1$ 等，用参数做分母需要考虑 0 分母情况。

3(4) 该题有很多种化简方法，此处给出一种很好的化简方法，见如下：

$$\text{解: } \begin{vmatrix} 5 & 0 & 4 & 2 \\ 1 & -1 & 2 & 1 \\ 4 & 1 & 2 & 0 \\ 1 & 1 & 1 & 1 \end{vmatrix} \xrightarrow{\substack{r_3+r_1 \\ r_4+r_2}} \begin{vmatrix} 5 & 0 & 4 & 2 \\ 1 & -1 & 2 & 1 \\ 5 & 0 & 4 & 1 \\ 2 & 0 & 3 & 2 \end{vmatrix} \xrightarrow{c_2 \text{ 展开}} - \begin{vmatrix} 5 & 4 & 2 \\ 5 & 4 & 1 \\ 2 & 3 & 2 \end{vmatrix} \xrightarrow{r_1-r_2} - \begin{vmatrix} 0 & 0 & 1 \\ 5 & 4 & 1 \\ 2 & 3 & 2 \end{vmatrix} = - \begin{vmatrix} 5 & 4 \\ 2 & 3 \end{vmatrix} = -7.$$

3(6) 很多同学直接展开，也可如下解答：

$$\text{解: } \begin{vmatrix} a & 0 & b & 0 \\ 0 & c & 0 & d \\ x & 0 & y & 0 \\ 0 & u & 0 & v \end{vmatrix} \xrightarrow{c_2 \leftrightarrow c_3} \begin{vmatrix} a & b & 0 & 0 \\ 0 & 0 & c & d \\ x & y & 0 & 0 \\ 0 & 0 & u & v \end{vmatrix} \xrightarrow{r_2 \leftrightarrow r_3} \begin{vmatrix} a & b & 0 & 0 \\ x & y & 0 & 0 \\ 0 & 0 & c & d \\ 0 & 0 & u & v \end{vmatrix} = a \begin{vmatrix} b & 0 & 0 \\ y & 0 & 0 \\ 0 & c & d \\ 0 & u & v \end{vmatrix} = \begin{vmatrix} a & b \\ x & y \end{vmatrix} \times \begin{vmatrix} c & d \\ u & v \end{vmatrix} = (ay-bx)(cv-du).$$

4(3) 有些同学利用行列式相减性质简化，可处理称范德蒙行列式再计算，见如下：

$$\text{解: } \begin{vmatrix} 1 & ax & a^2+x^2 \\ 1 & ay & a^2+y^2 \\ 1 & az & a^2+z^2 \end{vmatrix} \xrightarrow{c_3-a^2c_1} \begin{vmatrix} 1 & ax & x^2 \\ 1 & ay & y^2 \\ 1 & az & z^2 \end{vmatrix} \xrightarrow{c_3 \div a} \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \xrightarrow{\text{范德蒙}} a \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = a(y-x)(z-x)(z-y) = a(x-y)(y-z)(z-x).$$

4(4)有多种解答，见如下：

解法一：

$$(\text{列比例全加}) \begin{vmatrix} \lambda & -1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & -1 \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_2 & \lambda+a_1 \end{vmatrix} \xrightarrow{c_1+\sum_{i=2}^n \lambda^{-i-1}c_i} \begin{vmatrix} 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & -1 \\ \sum_{i=0}^n a_i \lambda^{n-i} & a_{n-1} & a_{n-2} & \cdots & a_2 & \lambda+a_1 \end{vmatrix} \xrightarrow{c_1 \text{ 展开}} (-1)^{n+1} \left(\sum_{i=0}^n a_i \lambda^{n-i} \right) \begin{vmatrix} -1 & & & & \\ & \lambda & -1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & -1 \end{vmatrix}_{n-1} = \sum_{i=0}^n a_i \lambda^{n-i}, \text{ 其中 } a_0=1.$$

解法二：

$$(\text{列比例递加}) \begin{vmatrix} \lambda & -1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & -1 \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_2 & \lambda+a_1 \end{vmatrix} \xrightarrow{c_{j-1}+\lambda c_j, j=n, n-1, \cdots, 2} \begin{vmatrix} 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & -1 \\ f_n(\lambda) & f_{n-1}(\lambda) & f_{n-2}(\lambda) & \cdots & f_2(\lambda) & \lambda+a_1 \end{vmatrix} = D_n, \text{ 其中 } f_k(\lambda) = \lambda^k + a_1 \lambda^{k-1} + \cdots + a_k.$$

$$D_n \xrightarrow{c_1 \text{ 展开}} (-1)^{n+1} f_n(\lambda) \begin{vmatrix} -1 & & & & \\ & -1 & & & \\ & & \ddots & \ddots & \\ & & & -1 & \\ & & & & -1 \end{vmatrix}_{n-1} = (-1)^{n+1} f_n(\lambda) (-1)^{n-1} = f_n(\lambda) = \lambda^n + a_1 \lambda^{n-1} + \cdots + a_n.$$

解法三：

$$(\text{第 1 列展开}) D_n = \begin{vmatrix} \lambda & -1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & -1 \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_2 & \lambda+a_1 \end{vmatrix} \xrightarrow{c_1 \text{ 展开}} \lambda D_{n-1} + (-1)^{n+1} a_n \begin{vmatrix} -1 & 0 & \cdots & 0 \\ \lambda & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 \end{vmatrix}_{n-1} = \lambda D_{n-1} + a_n.$$

$$\text{故有 } D_n = \lambda D_{n-1} + a_n = \lambda(\lambda D_{n-2} + a_{n-1}) + a_n = \lambda^2 D_{n-2} + a_{n-1} \lambda + a_n = \cdots = \lambda^{n-1}(\lambda + a_1) + a_2 \lambda^{n-2} + \cdots + a_{n-1} \lambda + a_n = \lambda^n + a_1 \lambda^{n-1} + \cdots + a_n.$$

解法四：

$$(\text{第 } n \text{ 行展开}) D_n = \begin{vmatrix} \lambda & -1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & -1 \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_2 & \lambda+a_1 \end{vmatrix} \xrightarrow{r_n \text{ 展开}} (-1)^{n+1} a_n M_{n1} + (-1)^{n+2} a_{n-1} M_{n2} + \cdots + (-1)^{n+j} a_{n-j+1} M_{nj} + \cdots + (-1)^{n+n} (\lambda+a_1) M_{nn},$$

其中

$$M_{nj} = \begin{vmatrix} \lambda & -1 & & \\ & \ddots & \ddots & \\ & & \lambda & -1 \\ & & & \lambda \\ -1 & & & \\ & \lambda & -1 & \\ & & \ddots & \ddots \\ & & & \lambda & -1 \end{vmatrix}_{(j-1)+(n-j)}$$

$$= \lambda^{j-1}(-1)^{n-j}, (-1)^{n+j} a_{n-j+1} M_{nj} = a_{n-j+1} \lambda^{j-1}, j=1, 2, \dots, n.$$

故 $D_n = a_n + a_{n-1}\lambda + \dots + a_{n-j+1}\lambda^{j-1} + \dots + (\lambda + a_1)\lambda^{n-1} = \lambda^{n-1} + a_1\lambda^{n-1} + \dots + a_{n-1}\lambda + a_n$.

解法五：数学归纳法，利用解法三的第一步 $D_n = \lambda D_{n-1} + a_n$.

5(2) 很多同学利用所有列加到第一列提取第一个因式，后面计算复杂，可利用分块特点计算如下：

解：

$$\begin{vmatrix} x & a & b & c \\ a & x & c & b \\ b & c & x & a \\ c & b & a & x \end{vmatrix} \xrightarrow[r_2+r_4]{r_1+r_3} \begin{vmatrix} x+b & a+c & x+b & a+c \\ a+c & x+b & a+c & x+b \\ b & c & x & a \\ c & b & a & x \end{vmatrix} \xrightarrow[c_4-c_2]{c_3-c_1} \begin{vmatrix} x+b & a+c & 0 & 0 \\ a+c & x+b & 0 & 0 \\ b & c & x-b & a-c \\ c & b & a-c & x-b \end{vmatrix}$$

$$= \begin{vmatrix} x+b & a+c \\ a+c & x+b \end{vmatrix} \begin{vmatrix} x-b & a-c \\ a-c & x-b \end{vmatrix} = ((x+b)^2 - (a+c)^2)((x-b)^2 - (a-c)^2) = 0,$$

故解为 $x+b = \pm(a+c), x-b = \pm(a-c)$ 即 $x = -b+a+c, x = -b-a-c, x = b+a-c, x = b-a+c$.

6(3) 计算不够简单，可如下：

解：

$$D = \begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} \xrightarrow[c_3-c_4]{c_1-c_2} \begin{vmatrix} x & 1 & 0 & 1 \\ x & 1-x & 0 & 1 \\ 0 & 1 & y & 1 \\ 0 & 1 & y & 1-y \end{vmatrix} \xrightarrow[c_3 \div y]{c_1 \div x} xy \begin{vmatrix} 1 & 1 & 0 & 1 \\ 1 & 1-x & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1-y \end{vmatrix} \xrightarrow[c_4-c_1-c_3]{c_2-c_1} xy \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & -x & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & -y \end{vmatrix} = x^2 y^2.$$

其余一些作业解题思路：

3(1) r_1, r_2 成比例值为 0 (2) $r_1 \leftrightarrow r_4, r_2 \leftrightarrow r_3$ 后计算 (3) c_1 或 r_1 展开 (5) $c_1+c_2+c_3+c_4, c_1 \div 10$ 计算

4(1) c_1-c_3, c_2-c_3, r_3 展开，或 $c_1-2c_2+c_3, c_1$ 展开；(2) $c_1+c_2+c_3, r_2-r_1, r_3-r_1$ ；(5) 参照补充例 1B

5(1) $c_1+c_2+c_3, r_2-r_1, r_3-r_1$ ；(3) 范德蒙展开，或参照补充例 1C

6(1) D 的第 3 行替换成 1,3,-2,2 然后计算；(2) $c_1-xc_2-yc_3-zc_4$